Practical 5

Ex.1 Exponential Distribution f(x) = (1/8) e^(−x/8), x > 0

Items	Value
Mean θ (f(x) = (1/θ) e^(−x/θ) for x > 0, else 0)	8
Rate λ = 1/θ	0.1250

*Find the probability	P()	value (x)	Equation	Answer	Check (EXPON.DIST)
(a) P(X ≤ 6)	P(X ≤ x)	6	P(X≤x) = 1 − e^(−x/θ)	**0.5276**	0.5276
(b) P(X ≤ 4)	P(X ≤ x)	4	P(X≤x) = 1 − e^(−x/θ)	**0.3935**	0.3935
(c) P(X ≥ 6)	P(X ≥ x)	6	P(X≥x) = e^(−x/θ)	**0.4724**	0.4724
(d) P(4 ≤ X ≤ 6)	x_1 (lower)	4	$P(x_1<X<x_2) = e^{(-x_1/\theta)} - e^{(-x_2/\theta)}$	**0.1342**	0.1342
	x_2 (upper)	6			

Ex.2 Web Page Download Time

Items	Value
Mean download time θ (seconds)	20
Rate λ = 1/θ	0.0500

*Find the probability	P()	value (x)	Equation	Answer	Check (EXPON.DIST)
(a) Less than 10 seconds	P(X < x)	10	P(X≤x) = 1 − e^(−x/θ)	**0.3935**	0.3935
(b) More than 30 seconds	P(X > x)	30	P(X≥x) = e^(−x/θ)	**0.2231**	0.2231
(c) Between 10 and 30 seconds	x_1 (lower)	10	$P(x_1<X<x_2) = e^{(-x_1/\theta)} - e^{(-x_2/\theta)}$	**0.3834**	0.3834
	x_2 (upper)	30			

Ex.3 Time Between Vehicle Arrivals

Items	Value
Mean time between arrivals θ (seconds)	12
Rate λ = 1/θ	0.0833

*Find the probability	P()	value (x)	Equation	Answer	Check (EXPON.DIST)
(a) 12 seconds or less	P(X ≤ x)	12	P(X≤x) = 1 − e^(−x/θ)	**0.6321**	0.6321
(b) 6 seconds or less	P(X ≤ x)	6	P(X≤x) = 1 − e^(−x/θ)	**0.3935**	0.3935
(c) 30 seconds or more	P(X ≥ x)	30	P(X≥x) = e^(−x/θ)	**0.0821**	0.0821

Ex.4 Gamma Distribution: Mean and Standard Deviation

Items	Value
Shape parameter α	2
Scale parameter β	3

*Find the mean and SD	Symbol	value	Equation	Answer
[1] Mean	μ		$\mu = \alpha \cdot \beta$	**6.0000**
[2] Variance	σ^2		$\sigma^2 = \alpha \cdot \beta^2$	**18.0000**
[3] Standard deviation	σ	18.0000	$\sigma = \sqrt{(\alpha \cdot \beta^2)} = \beta \cdot \sqrt{\alpha}$	**4.2426**

Ex.5 Gamma Distribution: P(X < 5)

Items (from Ex.4)	Value
Shape parameter α	2
Scale parameter β	3

*Find the probability	P()	value (x)	Equation	Answer	Check (GAMMA.DIST)
P(X < 5)	P(X < x)	5	$F(x) = 1 - e^{(-x/\beta)} \cdot (1 + x/\beta)$ (for α = 2)	**0.4963**	0.4963